Ritvik Singh

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Education

University of California Berkeley — Berkeley, USA
PhD Computer Science — 2025 - Present

- Advised by Pieter Abbeel and Jitendra Malik.

University of Toronto — Toronto, Canada
BASc Engineering Science - Machine Intelligence — 2019 - 2024

- GPA: 3.96/4.00

Employment

Amazon FAR Research Intern | San Francisco, USA — 2026.01 - Present

- Working on scaling robotics Behavior Cloning.

NVIDIA Deep Learning Engineer | Santa Clara, USA — 2024.06 - 2025.09

- Worked on end-to-end control for dexterous manipulation.

NVIDIA Deep Learning Engineering Intern | Toronto, Canada — 2023.05 - 2024.05

- Large-scale synthetic data generation for robotics pose estimation.
- 3D vision leveraging diffusion models and a custom differentiable PBR renderer for material generation.

NVIDIA Deep Learning Engineering Intern | Toronto, Canada — 2022.01 - 2022.12

- Scaled up synthetic data generation for in-hand manipulation.
- Worked on Omniverse Replicator and developing synthetic data pipelines for robotics and computer vision.

Publications

1. A. Allshire, H. Singh, **R. Singh** et al., Scalable Behavior Cloning with Open Data, Training, and Evaluation, *Preprint*
2. NVIDIA et al., Isaac Lab: A GPU-Accelerated Simulation Framework for Multi-Modal Robot Learning, *Preprint*
3. **R. Singh**, K. Van Wyk, J. Malik, P. Abbeel, N. Ratliff, and A. Handa, End-to-end RL Improves Dexterous Grasping Policies, *Preprint*
4. **R. Singh**, A. Allshire, A. Handa, N. Ratliff, and K. Van Wyk, DextrAH-RGB: Visuomotor Policies to Grasp Anything with Dexterous Hands, *ICRA 2025*
5. **R. Singh**, J.Liu, J. Lafleche, K. Van Wyk, Y. Chao, N. Ratliff, and A. Handa, Synthetica: Large Scale Synthetic Data Generation for Robot Perception, *IROS 2025*
6. A. Handa, A. Allshire, V. Makoviychuk, A. Petrenko, **R. Singh**, J. Liu, D. Makoviichuk, K. Van Wyk, A. Zhurkevich, B. Sundaralingam, Y. Narang, J. Lafleche, D. Fox, and G. State, DeXtreme: Transfer of Agile In-hand Manipulation from Simulation to Reality, *ICRA* 2023
7. M. Mittal, C. Yu, Q. Yu, J. Liu, N. Rudin, D. Hoeller, J. Lin Yuan, **R. Singh**, Y. Guo, H. Mazhar, A. Mandlekar, B. Babich, G. State, M. Hutter, and A. Garg, ORBIT: A Unified Simulation Framework for Interactive Robot Learning Environments, *RA-L* 2023
8. D. Turpin, T. Zhong, S. Zhang, G. Zhu, J. Liu, **R. Singh**, E. Heiden, M. Macklin, S. Tsogkas, S. Dickinson, and A. Garg, Fast-Grasp'D: Dexterous Multi-finger Grasp Generation Through Differentiable Simulation, *Arxiv*

Patents

1. Training machine learning models using simulation for robotics systems and applications, US18448049